Project Report

Project: Waste classification

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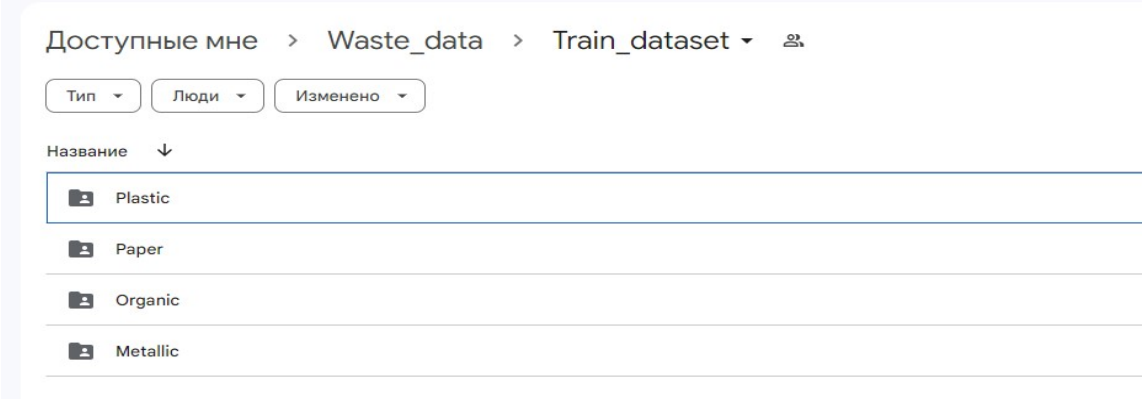
Background: Waste management has recently become a significant problem globally. According to statistics, every person daily produces around 0.75kg of waste. Most of the wastes don't go through recycling processing and end up being in landfills. This results in several issues including land, water and air pollution, increase in landfills, and land toxification. Therefore, the improved waste management and recycling policies are required. In this project, a waste classification model is proposed to facilitate automatic waste sorting using IoT and machine learning.

Purpose: The project is aimed to train image classification model for waste sorting. In the University campus, there are 4 different Dustbins for 4 different types of waste. Our project focuses mainly on those 4 types, which will be discussed in the later section of the report. Our ultimate purpose for this project is to integrate this machine learning model with IoT and fully automate the process of waste sorting around the campus. Later, this model can also be implemented in the Hotel Icon for the same purpose. This model can make the waste sorting process efficient, easy, and quick.

Data collection and Data Preparation: For the project we collected images from the internet. Then we divided the images into 4 classes: metallic, organic, paper and plastic. For each class we distributed approximately 100 images and divided them into Test Data (20% of images) and Training Data (80% of images).



Sample Images:

For sample images, we uploaded following type of images for different waste categories:

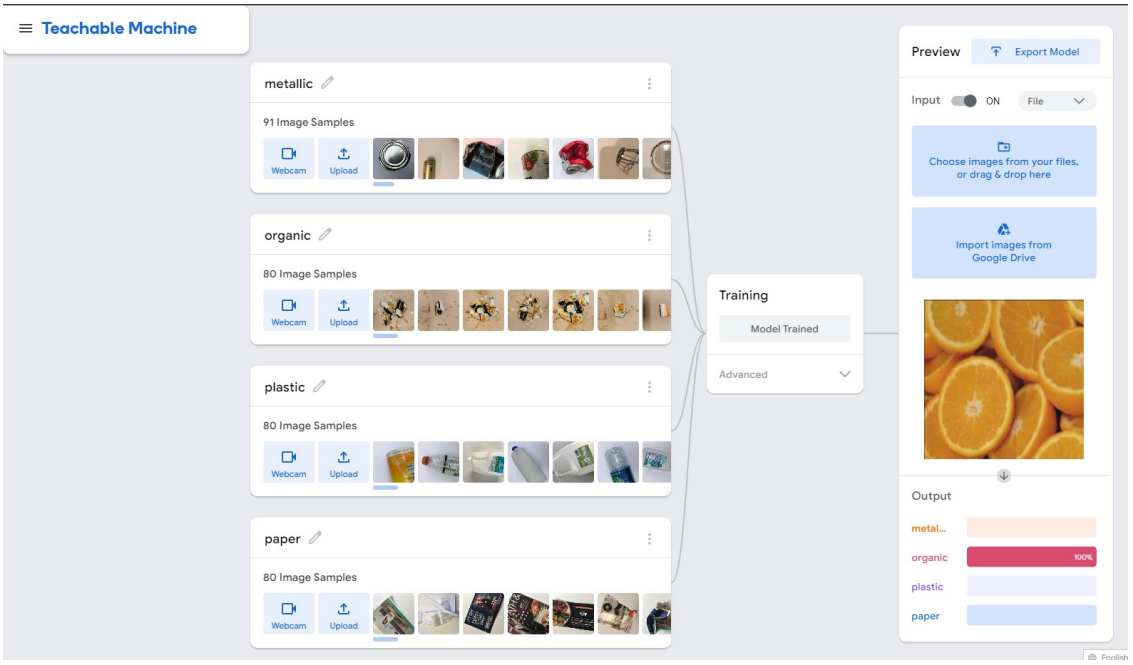
Organic:

Plastic

Metallic:

Paper:

Training the model: After collecting the data, we uploaded images to Teachable Machine. It is a web-based tool that makes model training easier and faster.



Real-time testing:

Output

metal...	87%
organic	0%
plastic	
paper	

Output

metal...	
organic	
plastic	
paper	99%

Output

metal...	
organic	90%
plastic	
paper	0%

Output

metal...	
organic	14%
plastic	75%
paper	0%

Output

metal...	100%
organic	
plastic	
paper	

Output

metal...	
organic	100%
plastic	
paper	

Limitations: As evident from the real_time test images, the model does not recognize the background wastes but only focuses on a single type of waste in a single run. In the image number 5 above, the model does not recognize the paper as well as plastic in the dustbin rather it only detects metal (monster can).

Conclusions: It seems that the model performs well with high accuracy in real-time data. For further improvement, we can train an object detection model that can detect more types of wastes in a single image.